

IE3010 Network Programming

NetMessenger: Multi-Client Chat and File-Sharing Platform over TCP/IP

Registration Number	IT23861404
Numeric Part	23861404
Last Four Digits	1404
TCP Port	7404
NID	8614
Server Source	server_1404.c
Client Source	client_1404.c
Makefile	Makefile_1404
Log File	netmsg_IT23861404.log
Storage	./storage/IT23861404/
ZIP	IE3010_IT23861404.zip

1. Introduction

NetMessenger is a TCP/IP client-server chat and file-sharing application implemented in C using the BSD socket API. The server accepts multiple clients concurrently. Registered clients can broadcast messages, send private messages, use rooms, list users and rooms, and transfer files through the server.

2. Personalisation Calculation

The numeric registration part is 23861404. The last four digits are 1404, so the listening port is $6000 + 1404 = 7404$. Digits 3–6 are 8614, so the response tag is NID:8614. The personalised source, log, storage and ZIP names are listed above.

3. Architecture and Concurrency

The system consists of one TCP server and multiple TCP clients. The server listens on TCP port 7404 and creates a POSIX thread for each connected client. Shared user and room state is protected by a mutex. This allows one client to wait for input without blocking other clients.

4. Protocol Implementation

The implementation supports REGISTER, LIST, BCAST, PMSG, JOIN, LEAVE, ROOMS, RMSG, SENDFILE and QUIT. Text commands are newline terminated. Server OK and ERR responses contain NID:8614. SENDFILE is read using the exact declared byte count.

5. Receiver-side File Framing Assumption

The brief defines the client-to-server SENDFILE framing and raw bytes but does not explicitly define receiver-side notification framing. This implementation uses FILE followed immediately by exactly raw bytes. This assumption should be explained during the viva.

6. Functional Features

Registration enforces unique usernames. LIST returns connected users. BCAST delivers messages to other clients. PMSG targets one named client. JOIN and LEAVE manage rooms, ROOMS lists rooms, and RMSG delivers room messages. SENDFILE stores a server-side copy and forwards the file to a user or room. Disconnect cleanup removes users from rooms.

7. Error Handling

The server returns structured errors for missing registration, duplicate usernames, invalid commands, unknown users/rooms, oversized or invalid files, and storage failures. SIGPIPE is ignored so a disconnected peer does not terminate the server process.

8. Testing Summary

Test	Observed result
Server start	Listening on TCP 7404
Registration	OK REGISTERED with NID:8614
LIST	Connected usernames returned
BCAST	Other client received message
PMSG	Target received private message
JOIN / RMSG / ROOMS	Room operations worked
SENDERFILE	2066-byte test file forwarded; SHA-256 matched
Invalid command	ERR 005 INVALID_COMMAND NID:8614
QUIT	OK BYE NID:8614

9. Evidence to Insert Before Submission

Insert genuine screenshots from your own Linux environment showing the server on port 7404, ss -tlnp output, multiple clients, NID:8614 responses, broadcast/private/room messaging, successful file transfer, personalised storage, the log, an error response, and an unexpected disconnect. Do not fabricate or edit screenshots.

10. Conclusion

The implementation provides the mandatory NetMessenger functionality and applies the IT23861404 personalisation. The supplied test run verified registration, users, broadcast, private messaging, rooms, file transfer, error handling and clean quit. The program should still be compiled and demonstrated on the department Linux environment before final submission.